DHAYALAN K

Full-Stack Software Engineer | React, Node.js, Express, Java OOP, MongoDB and Systems

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IMPACT AT A GLANCE

9 Shipped Projects including full e-commerce platforms, community forum extensions & socket protocols | **92% Merit Score** in Java OOP & MongoDB Data Modeling | **AWS ECS & Cloud Certified** Amazon ECS Knowledge Badge & CI/CD CDK Pipelines | **3 National CTF Challenges** Tamil Nadu Police Yukthi CTF 2.0, Amrita Lemon CTF, REC HackTiVate

Full-Stack Software Engineer with proven engineering expertise across frontend systems (React, Tailwind CSS, Mithril.js), high-throughput backend APIs (Node.js, Express, Java, Python), and database optimization (MongoDB, MySQL). Built production-grade web platforms with client-side state persistence, custom modular UI extension engines, and containerized microservices. Accelerated web load speeds by 40% through decoupled architectures and optimized query indexing.

ENGINEERING HIGHLIGHTS & ARCHITECTURE

Production Web Systems & Platform Engineering

- Engineered **Shahihennaholic Storefront**: mobile-first e-commerce catalog featuring client-side shopping cart calculations and zero-friction LocalStorage state synchronization.
- Designed dynamic bridal catalog views with instantaneous page responsiveness, automated cart state serialization, and direct WhatsApp order payload dispatch.
- Extended **Flarum Community Platform** by architecting **FLARUM1**, an independent companion JavaScript extension hooking into Mithril.js single-page application lifecycles.
- Tuned MySQL schema design, query caching, and relational indexes for high-frequency discussion threads across concurrent user sessions without core upstream modifications.
- Architected **Java ATM Banking Engine (maincode.java)**: strict OOP encapsulation model with PIN authentication, transaction rollback protection, and formatted audit statement ledgers.

Real-Time Telemetry & Systems Integration

- Constructed **Browser Monitoring & Telemetry Daemon** in Python: instrumented live browser sessions, capturing DOM mutations, event listeners, and network interactions.
- Designed fault-tolerant local JSON buffer queue guaranteeing zero event loss during network interruptions, paired with streaming reporting pipelines.
- Containerized full-stack services with Docker and Docker Compose, achieving reproducible local environments and zero-downtime rolling restart workflows.

TECHNICAL SKILLS

Frontend Engineering: React (v18+), Hooks (useState, useEffect, useCallback, useMemo), Context API, JavaScript (ES6+), TypeScript, Next.js, Tailwind CSS, HTML5, CSS3, DOM Telemetry, Mithril.js

Backend & API Architecture: Node.js, Express.js, Java (OOP, Multi-threading, Collections), Python, RESTful API Design, JWT Authentication, CORS, Middleware Pipelines, Asynchronous Programming

Databases & Storage: MongoDB (Mongoose ODM, Schema Validation, Aggregation Pipelines, Indexing — 92% Score), MySQL, LocalStorage State, In-Memory Caching

DevOps, Cloud & Tooling: Docker, Docker Compose, Amazon Web Services (ECS, CDK, CloudFormation), Git, GitHub Actions, Linux (Ubuntu/Debian), Nginx, Postman, Vite, NPM/Yarn

Frameworks & Standards: Component-Driven Architecture, Microservices, MVC Pattern, REST API Specifications, Responsive Web Design

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer, Web Platforms & Open Source Systems **2024 to Present, Coimbatore**

- Architected responsive, database-backed web applications using React, Node.js, Express, MongoDB, and modern utility-first styling systems.
- Developed client-side state persistence workflows with LocalStorage and real-time DOM mutation monitoring for consumer web applications.
- Designed RESTful APIs with input validation, JWT token verification, and structured error-handling middleware.

Systems & Automation Engineer, Independent Systems Lab **2025 to Present, Coimbatore**

- Containerized multi-tier full-stack workloads using Docker, standardizing environment parity across development and deployment stages.
- Automated database backup pipelines, directory organization daemons, and system health checks in Python and Bash.

KEY PROJECTS & CODEBASES

Shahihennaholic E-Commerce & Ordering Platform github.com/DHAYALANKANIAPPAN/PROJECTS

- Built full responsive product showcase and shopping cart with client-side price computation and order serialization.
- Streamlined checkout workflow into automated WhatsApp messaging payloads, reducing order drop-off. Stack: HTML5, CSS3, JavaScript, LocalStorage.

Flarum Discussion Forum & FLARUM1 Extension github.com/DHAYALANKANIAPPAN/flarum

- Authored decoupled JavaScript frontend extension hooking into Mithril.js component mounts for custom themes and controls.
- Configured MySQL index optimizations for fast discussion lookup under load. Stack: PHP, JavaScript, Mithril.js, MySQL.

ATM Banking & Account Management System github.com/DHAYALANKANIAPPAN/PROJECTS

- Engineered Java OOP banking system implementing encapsulation, account authorization, and atomic balance mutation safeguards. Stack: Java, OOP, CLI.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering (Cyber Security), Pursuing

Sri Shakthi Institute of Engineering and Technology, Coimbatore (2024 – 2028) | Grade Point Average: 8.21 out of 10.0.
Available for full-time employment & internships.

VERIFIED CERTIFICATIONS, AWARDS & ACTIVITIES

- Certificate of Merit: Java + Mongo DB (92% Score) — Hazhtech Career Development Centre / IACT (60-day OOP & NoSQL training).
- Amazon ECS Knowledge Badge Assessment — Amazon Web Services (AWS) · Credly Official Credential.
- Deploying a CI/CD Pipeline with AWS CDK — AWS Skill Builder Official Lab Certification.
- Yukthi CTF 2.0 Participant — National-Level Cyber Defense & Exploitation Event, Tamil Nadu Police & Selfmade Ninja Academy.
- 0xTi - HackTiVate 24-Hour CTF Participant — Rajalakshmi Engineering College (Intensive software problem-solving).